

Metric Entropy and Observational Entropy Defect

Covering Numbers, Hidden Fibers, Chromatic Graphs, and Heat-Trace Regularization

Stassis Research Program
Geometry of Observation note

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Abstract

This note formulates a strict metric-entropy layer for Geometry of Observation. The basic question is not only how far points are, but how many distinguishable metric cells are needed to observe a space at resolution ε . Covering numbers $N_\varepsilon(X, d)$ and metric entropy $H_\varepsilon = \log N_\varepsilon$ provide a scale-dependent measure of distinguishability. An observation map $\mathcal{O} : X \rightarrow Y$ may collapse hidden fibers; the difference between hidden and observed covering entropy is defined as an observational entropy defect. We prove elementary contraction statements for Lipschitz observations, compute the asymptotic defect of hidden compact fibers such as S^1 , record the finite defect of half-step parity aliasing, define chromatic entropy for lifted color graphs, and connect local covering capacity with curvature through the small-ball volume expansion and heat-trace asymptotics. The note is a mathematical interface only. It does not derive gravity, prove a new entropy law, or replace information theory, optimal transport, or spectral geometry.

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1 Scope and claim firewall

The present document adds a metric-entropy module to the Geometry of Observation program. The guiding idea is

observation loses not merely coordinates, but distinguishability at a chosen scale.

The central objects are covering numbers, entropy defects under observation maps, and scale-dependent color or feature complexity.

Boundary. The note does not claim that metric entropy is physical mass, that curvature is caused by color, or that covering numbers replace Einstein geometry. The permitted claim is narrower: covering entropy gives a rigorous language for the amount of distinguishable structure retained or lost under observation.

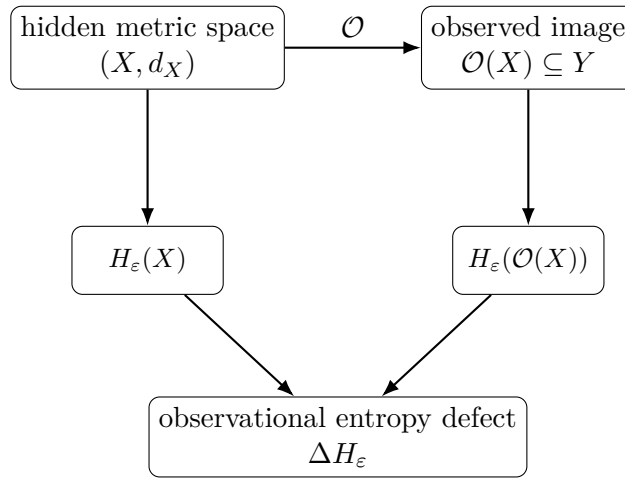


Figure 1: Observation as a loss or transformation of scale-dependent distinguishability.

2 Covering numbers and metric entropy

Definition 2.1 (Covering number). Let (X, d) be a metric space and $A \subseteq X$. For $\varepsilon > 0$, the covering number is

$$N_\varepsilon(A, d) = \min \left\{ N : A \subseteq \bigcup_{j=1}^N B_d(x_j, \varepsilon) \right\},$$

when such a finite cover exists. If no finite cover exists, set $N_\varepsilon(A, d) = +\infty$.

Definition 2.2 (Metric entropy). The metric entropy of A at scale ε is

$$H_\varepsilon(A, d) = \log N_\varepsilon(A, d).$$

It is a scale-dependent measure of distinguishability, not a thermodynamic entropy by itself.

Definition 2.3 (Packing number). The packing number $P_\varepsilon(A, d)$ is the maximal cardinality of a subset $\{a_1, \dots, a_N\} \subseteq A$ satisfying

$$d(a_i, a_j) > \varepsilon \quad (i \neq j).$$

Packing counts how many mutually distinguishable samples can be placed inside A at separation ε .

Proposition 2.4 (Basic covering-packing bounds). *For every metric space and $A \subseteq X$,*

$$P_{2\varepsilon}(A, d) \leq N_\varepsilon(A, d) \leq P_\varepsilon(A, d)$$

whenever the right-hand packing number is finite.

Proof. If A is covered by ε -balls, then a 2ε -separated set contains at most one point in each ball; hence $P_{2\varepsilon} \leq N_\varepsilon$. For the second inequality, take a maximal ε -separated set. Maximality implies that the corresponding ε -balls cover A . $\square$

Definition 2.5 (Entropy dimension). When the limit exists, define

$$\dim_{\text{ent}}(A, d) = \lim_{\varepsilon \rightarrow 0} \frac{H_\varepsilon(A, d)}{\log(1/\varepsilon)}.$$

This is the usual box-counting dimension when expressed through covering numbers.

Interpretation. Dimension can be read as the rate at which distinguishable cells proliferate as resolution increases:

$$\text{dimension} = \text{growth rate of distinguishability under } \varepsilon \downarrow 0.$$

3 Lipschitz observations and entropy contraction

Definition 3.1 (Observation map). An observation map is a map between metric spaces

$$\mathcal{O} : (X, d_X) \rightarrow (Y, d_Y).$$

It may be injective, non-injective, Lipschitz, smooth, singular, or a quotient map. The present section treats the Lipschitz case.

Theorem 3.2 (Lipschitz observation entropy bound). *Let $\mathcal{O} : X \rightarrow Y$ be L -Lipschitz:*

$$d_Y(\mathcal{O}(x), \mathcal{O}(x')) \leq L d_X(x, x').$$

Then for every $A \subseteq X$ and $\varepsilon > 0$,

$$N_\varepsilon(\mathcal{O}(A), d_Y) \leq N_{\varepsilon/L}(A, d_X),$$

with the convention that $\varepsilon/L = +\infty$ if $L = 0$. Equivalently,

$$H_\varepsilon(\mathcal{O}(A), d_Y) \leq H_{\varepsilon/L}(A, d_X).$$

Proof. Let $A \subseteq \bigcup_{j=1}^N B_X(x_j, \varepsilon/L)$. If $x \in B_X(x_j, \varepsilon/L)$, then

$$d_Y(\mathcal{O}(x), \mathcal{O}(x_j)) \leq L d_X(x, x_j) < \varepsilon.$$

Thus

$$\mathcal{O}(A) \subseteq \bigcup_{j=1}^N B_Y(\mathcal{O}(x_j), \varepsilon),$$

so the same number of centers covers the observed image at scale ε . $\square$

Definition 3.3 (Scale-corrected observational entropy defect). For an L -Lipschitz observation map define

$$\mathcal{I}_\mathcal{O}^L(A; \varepsilon) := H_{\varepsilon/L}(A, d_X) - H_\varepsilon(\mathcal{O}(A), d_Y) \geq 0.$$

This nonnegative number measures hidden distinguishability lost or collapsed by the observation at matched scale.

Remark 3.4 (Uncorrected defect). The uncorrected expression

$$H_\varepsilon(A, d_X) - H_\varepsilon(\mathcal{O}(A), d_Y)$$

is useful as a diagnostic only after a common scale convention has been fixed. It need not be nonnegative for arbitrary metrics and non-unit Lipschitz constants.

4 Hidden fibers and entropy loss

Definition 4.1 (Projection fiber). Let $E = B \times F$ with the max product metric

$$d_E((b, f), (b', f')) = \max\{d_B(b, b'), d_F(f, f')\}.$$

The projection

$$\pi : E \rightarrow B, \quad \pi(b, f) = b,$$

forgets the fiber coordinate.

Proposition 4.2 (Product upper bound). *For compact metric spaces B and F ,*

$$N_\varepsilon(B \times F, d_E) \leq N_\varepsilon(B, d_B) N_\varepsilon(F, d_F).$$

Consequently,

$$H_\varepsilon(B \times F) \leq H_\varepsilon(B) + H_\varepsilon(F).$$

Proof. Take ε -covers $B \subseteq \bigcup_i B_B(b_i, \varepsilon)$ and $F \subseteq \bigcup_j B_F(f_j, \varepsilon)$. Then the product balls centered at (b_i, f_j) cover $B \times F$ in the max metric. $\square$

Proposition 4.3 (Compact-manifold fiber defect). *Let B and F be compact Riemannian manifolds of dimensions m and n . For the projection $\pi : B \times F \rightarrow B$, the hidden-fiber contribution satisfies*

$$H_\varepsilon(B \times F) - H_\varepsilon(B) = n \log(1/\varepsilon) + O(1)$$

as $\varepsilon \rightarrow 0$.

Proof sketch. For a compact k -dimensional Riemannian manifold M , standard small-scale volume comparison gives

$$N_\varepsilon(M) = C_M \varepsilon^{-k} (1 + o(1))$$

up to multiplicative constants sufficient for entropy asymptotics. Taking logarithms gives

$$H_\varepsilon(M) = k \log(1/\varepsilon) + O(1).$$

Apply this to $B \times F$ and to B . $\square$

Corollary 4.4 (Hidden Kaluza-Klein circle). *If $E = B \times S^1$ and $\pi : E \rightarrow B$ forgets the compact circle, then*

$$H_\varepsilon(E) - H_\varepsilon(B) = \log(1/\varepsilon) + O(1).$$

One hidden compact circle contributes one logarithmic unit of small-scale entropy growth.

Corollary 4.5 (Half-step parity aliasing). *Let the hidden fiber be the two-point parity space $F = \mathbb{Z}/2\mathbb{Z}$ with discrete metric. For ε smaller than the point separation,*

$$H_\varepsilon(F) = \log 2.$$

Thus an observation that forgets the parity state loses exactly $\log 2$ of finite fiber entropy at resolving scales.

5 Cylindrical observation fibers

The cylindrical observation model considered in the companion note has a schematic parameter space

$$\mathcal{P} = S^2 \times D \times S^1 \times SO(2),$$

where S^2 records axis direction, $D \subset \frac{1}{2}\mathbb{N}$ records diameter levels, S^1 records step or phase, and $SO(2)$ records cross-section orientation.

If an observation map collapses the hidden cylindrical state to a one-dimensional trace,

$$\mathcal{O} : \mathcal{P} \rightarrow \mathbb{R},$$

then the observed entropy may grow only as

$$H_\varepsilon(\mathcal{O}(\mathcal{P})) \sim \log(1/\varepsilon),$$

while the hidden parameter space can carry contributions from S^2 , S^1 , $SO(2)$, and the discrete diameter ladder.

Interpretation. A simple visible point trace may hide a high-entropy fiber of directions, phases, orientations, and diameter states. This is the entropy-theoretic version of the statement that observation can create false simplicity.

6 Chromatic metric entropy

Definition 6.1 (Chromatic field). Let (X, d_X) be a metric space and let $(\mathcal{C}, d_{\mathcal{C}})$ be a color, phase, feature, or label space. A chromatic field is a map

$$c : X \rightarrow \mathcal{C}.$$

Definition 6.2 (Graph lift and chromatic metric). The graph of c is

$$\Gamma_c = \{(x, c(x)) : x \in X\} \subseteq X \times \mathcal{C}.$$

For $\lambda > 0$, define the product metric

$$D_\lambda((x, a), (y, b))^2 = d_X(x, y)^2 + \lambda^2 d_{\mathcal{C}}(a, b)^2.$$

The chromatic metric entropy is

$$H_\varepsilon^{\text{ch}}(X, c; \lambda) = \log N_\varepsilon(\Gamma_c, D_\lambda).$$

Proposition 6.3 (Lipschitz chromatic upper bound). *If $c : X \rightarrow \mathcal{C}$ is L_c -Lipschitz, then the graph map*

$$\iota_c : X \rightarrow X \times \mathcal{C}, \quad \iota_c(x) = (x, c(x)),$$

is $\sqrt{1 + \lambda^2 L_c^2}$ -Lipschitz. Hence

$$H_\varepsilon^{\text{ch}}(X, c; \lambda) \leq H_{\varepsilon/\sqrt{1 + \lambda^2 L_c^2}}(X, d_X).$$

Proof. For $x, y \in X$,

$$D_\lambda(\iota_c(x), \iota_c(y))^2 = d_X(x, y)^2 + \lambda^2 d_{\mathcal{C}}(c(x), c(y))^2 \leq (1 + \lambda^2 L_c^2) d_X(x, y)^2.$$

Apply Definition 3.2. □

Definition 6.4 (Chromatic entropy load). The excess distinguishability load of the chromatic field is the diagnostic

$$\mathcal{L}_\varepsilon^{\text{ch}}(X, c; \lambda) = H_\varepsilon^{\text{ch}}(X, c; \lambda) - H_\varepsilon(X, d_X),$$

whenever the same metric scale convention is intended.

7 Local entropy pressure and curvature

Let (M, g) be a compact Riemannian n -manifold and let $c : M \rightarrow \mathcal{C}$ be a chromatic field. For a radius $R > 0$, define a local entropy density

$$\rho_{\varepsilon, R}^{\text{ch}}(x) = \frac{H_{\varepsilon}^{\text{ch}}(B_g(x, R), c; \lambda)}{\text{Vol}_g(B_g(x, R))}.$$

This is not physical mass. It is a scale-dependent density of distinguishability.

Proposition 7.1 (Curvature and local covering capacity). *For a smooth Riemannian n -manifold, the small geodesic ball volume satisfies*

$$\text{Vol}_g(B_g(x, r)) = \omega_n r^n \left[1 - \frac{\text{Scal}_g(x)}{6(n+2)} r^2 + O(r^4) \right]$$

as $r \rightarrow 0$.

Interpretation. Curvature changes the local capacity of metric cells. Positive scalar curvature decreases the volume of small balls relative to the Euclidean model, while negative scalar curvature increases it. Therefore any variational model using local covering capacity can couple naturally to curvature, without yet being a theory of gravity.

Definition 7.2 (Entropy-pressure functional). A schematic entropy-pressure term is

$$\mathcal{P}_{\varepsilon, R}[g, c] = \int_M \Phi(\rho_{\varepsilon, R}^{\text{ch}}(x)) \, dV_g(x),$$

where Φ is a nonnegative increasing penalty function. A full variational model may couple this to curvature terms, for example

$$\mathcal{J}[g, c] = \alpha \int_M R_g \, dV_g + \beta \int_M |\nabla_g c|^2 \, dV_g + \gamma \mathcal{P}_{\varepsilon, R}[g, c].$$

Boundary. The last display is a mathematical ansatz. It is not a derivation of the Einstein equation. A physical claim would require units, a stress tensor, a variational theory, calibration, and comparison with known gravitational data.

8 Heat trace as smooth spectral entropy

Covering numbers are often nondifferentiable with respect to geometric data. A smoother spectral proxy is the heat trace.

Definition 8.1 (Heat trace). For a compact Riemannian manifold (M, g) with nonnegative Laplace-Beltrami operator Δ_g , define

$$Z_g(t) = \text{Tr}(e^{-t\Delta_g}) = \sum_{j=0}^{\infty} e^{-t\lambda_j}, \quad t > 0.$$

A spectral entropy proxy is

$$H_t^{\text{heat}}(M, g) = \log Z_g(t).$$

Proposition 8.2 (Small-time heat trace expansion). *As $t \downarrow 0$,*

$$Z_g(t) \sim (4\pi t)^{-n/2} \left[\text{Vol}_g(M) + \frac{t}{6} \int_M \text{Scal}_g \, dV_g + O(t^2) \right].$$

Interpretation. The heat trace is a smooth spectral version of scale-dependent distinguishability. It is also the bridge to theta functions in compact spectral sectors: on tori and lattices, heat traces are theta sums.

9 Unit distances, shells, and metric entropy

Given a finite set $P = \{p_1, \dots, p_N\}$ in a metric space, the radius- r shell graph is

$$G_{r,\sigma}(P) = \{(i, j) : |d(p_i, p_j) - r| < \sigma\}.$$

The classical unit-distance graph is the limiting exact shell at $r = 1$ in Euclidean space.

Covering and packing numbers do not solve the unit-distance problem, but they control neighboring questions: how many points can be placed with a given separation, how many local shells can be populated, and how much chromatic conflict arises when nearby or shell-adjacent points require distinct labels.

Interpretation. The unit-distance problem counts exact metric shell coincidences. Metric entropy counts distinguishable cells. Chromatic metric stress couples the two: shell adjacency becomes a local constraint, and covering entropy measures how many distinguishable states the metric is trying to carry.

10 Research program

The metric-entropy module suggests the following precise tasks:

1. Compute observational entropy defects for projections $X \times F \rightarrow X$ for compact and discrete fibers.
2. Compare covering entropy, heat-trace entropy, and chromatic graph entropy on the same examples.
3. Define entropy-pressure stress tensors by varying smooth entropy proxies with respect to the metric.
4. Study unit-distance shell graphs under observation maps and compare their chromatic entropy to classical covering and packing numbers.
5. Develop a recoverability criterion: which entropy scales of X can be reconstructed from $\mathcal{O}(X)$ plus auxiliary measurements?

11 Claim audit

Allowed statement	Not allowed without further proof
H_ε measures scale-dependent distinguishability.	Metric entropy is physical entropy in every context.
Lipschitz observations satisfy covering-entropy bounds.	Observation maps preserve all geometric information.
Hidden compact fibers add logarithmic covering entropy.	A hidden fiber is automatically a physical compact dimension.
Chromatic entropy measures the covering complexity of a graph lift.	Color is physical mass or charge by definition.
Curvature affects local volume and covering capacity.	Entropy pressure derives Einstein gravity.
Heat traces provide smooth spectral entropy proxies.	Theta functions or heat traces are particle spectra.

12 Conclusion

Metric entropy supplies a concise mathematical language for Geometry of Observation:

hidden space $\rightarrow$ covering entropy $\rightarrow$ observation map $\rightarrow$ entropy defect.

The strongest invariant introduced here is

$$\mathcal{I}_{\mathcal{O}}^L(A; \varepsilon) = H_{\varepsilon/L}(A) - H_{\varepsilon}(\mathcal{O}(A)).$$

It measures hidden distinguishability lost at a chosen observation scale. This does not replace metric geometry; it complements it by asking how much distinguishable structure survives a readout.